Robot Status

Recorded condition and recommended next steps

Review details	
Client	Synthetic demonstration
Robot serial	Not available in supplied evidence
Robot model	Not established
Controller software	Not established
Incident dates	2026-04-01
Report generated	2026-11-02 18:18 UTC

Attention required

Controller evidence identifies concerns involving Base. These are the first areas to investigate; the evidence does not establish that the joints need replacement.

Start with the recorded process stage, physical clearance and active load settings. Use the joint summaries to distinguish application-related warnings from controller messages that require service interpretation.

Evidence is insufficient to characterize Shoulder, Elbow, Wrist 1, Wrist 2, Wrist 3. These joints remain unassessed or only partly observed; absence of a recorded fault is not a clean bill of health.

This assessment describes the available records. It is not an inspection of the robot’s present mechanical condition or a safety approval.

Joint condition

Each joint is assessed separately. “Not assessed” and “Limited evidence” are gaps in the available records, not a pass or a prediction of remaining life.

Assessment by joint	
Base	Needs attention The controller recorded collision detection or path-following warnings for Base. No matched joint telemetry was available to corroborate the warnings. These findings do not establish that the joint is damaged.
Shoulder	Limited evidence No mapped warning naming Shoulder was found in the supplied logs. Without usable joint telemetry or inspection evidence, its mechanical condition remains unassessed.
Elbow	Limited evidence No mapped warning naming Elbow was found in the supplied logs. Without usable joint telemetry or inspection evidence, its mechanical condition remains unassessed.
Wrist 1	Limited evidence No mapped warning naming Wrist 1 was found in the supplied logs. Without usable joint telemetry or inspection evidence, its mechanical condition remains unassessed.
Wrist 2	Limited evidence No mapped warning naming Wrist 2 was found in the supplied logs. Without usable joint telemetry or inspection evidence, its mechanical condition remains unassessed.
Wrist 3	Limited evidence No mapped warning naming Wrist 3 was found in the supplied logs. Without usable joint telemetry or inspection evidence, its mechanical condition remains unassessed.

Recommendations below connect the observed concerns to practical checks. Full incident measurements can be included in the optional technical evidence appendix.

Local evidence-based review; hypotheses require verification. Controller detections do not alone prove physical contact or component failure. Not a safety validation.

Program areas to review

Approach path and clearance

Program context	
Program	Program not identified
Review priority	Investigate
Evidence association	No matching captured program context

The controller reported collision detection or a path-following discrepancy. The approach, retract path and external cabling are relevant review areas, together with the active load model.

Supporting cases: INC-001

Blend configuration and waypoint continuity

Program context	
Program	Program not identified
Review priority	Investigate
Evidence association	No matching captured program context

The controller reported a blend issue. Use the full subtype: some C171 messages report a blend that is too small, while others report skipped waypoints.

Supporting cases: INC-001

Recommended action plan

Recommended next steps

1. Review Base at the affected motion stage: check physical clearance, active payload and centre of gravity, then acceleration changes and move transitions.

2. Inspect the real cell at the recorded approach/retract stage and compare fixture, workpiece and cable clearance with the intended path.

3. Obtain the matching flight recording, captured script and installation before assigning this concern to a specific program area or line.

4. Review the named waypoint, neighboring radii and executed branch using the complete C171 message. Adjust radius or spacing only after checking the required process path and clearance.

5. Collect representative controller logs and joint motion recordings for the insufficiently covered joints, and confirm current condition with the maintenance history and physical inspection.

Verify the application and physical setup before changing motion or load settings. Retain the original evidence for follow-up and service review.

Scope of this assessment

Reviewed 1 controller log dataset(s) and 0 flight recording(s).

The findings describe the supplied history and recorded windows, not continuous monitoring or a physical inspection.

Available incident dates: 2026-04-01.

Evidence still needed

This review does not predict remaining joint life or certify that the robot is safe to operate. Findings must be considered with the real application and any maintenance completed after these records.

Robot identity was not established from source serial numbers. Confirm all supplied files belong to the robot named on this report.

Recommended action plan (continued)

Technical references

1. UR technical guidance
https://www.universal-robots.com/articles/ur/robot-care-maintenance/understanding-protective-stops/

2. UR technical guidance
https://www.universal-robots.com/manuals/EN/HTML/SW5_24/Content/prod-err-codes/topics/CODE_153.html

3. UR technical guidance
https://www.universal-robots.com/manuals/EN/HTML/SW5_24/Content/prod-err-codes/topics/CODE_171.html

Local evidence-based review; hypotheses require verification. Controller detections do not alone prove physical contact or component failure. Not a safety validation.

Physical installation follow-up

Not assessed by this report. Source files and telemetry cannot verify these items. Record the inspection evidence and outstanding actions in the deployment checklist.

Manual checks · all unassessed	
Mounting and orientation	Record the base/tool mounting assessment, fastening inspection and configured orientation. Include moving-base conditions where applicable. Source: Guide p. 5; Workshop pp. 5–10
Environment and protection	Record suitability for the arm, controller and pendant: contamination, moisture, temperature, ventilation and cable routing. Source: Guide p. 5; Workshop pp. 12–18
Power and I/O	Record the electrical inspection reference, power quality and I/O load review. Use model-specific electrical limits. Source: Guide p. 9; Workshop p. 17
Risk assessment	Reference the application risk assessment and approved safety configuration, including tooling, workpieces and connected machinery. Do not store passwords here. Source: Guide p. 12
Safety validation	Reference the qualified validation record for emergency stops, safeguards, modes and connected safety I/O after installation or modification. Source: UR e-Series Service Manual, Pre-Use Assessment (SW5.24)
Inspection plan	Record the model-specific inspection plan, responsible person and next review date. Use the manual and operating conditions to set intervals. Source: Workshop pp. 54–56; UR e-Series Service Manual, Inspection Plan (SW5.24)

Guide: UR Installation Guide v1.3.3. Workshop: UR Application Deployment Workshop, May 2024. Page numbers refer to the supplied PDFs. Apply the current model-specific manual; these prompts are not a complete safety assessment.